

Sales Performance Analysis

Alfido Tech – Data Analyst Internship

“Analyzing sales performance across products, categories, regions and time to identify meaningful business insights.”

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Role	Data Analyst Intern
Organization	Alfido Tech
Project	Sales Performance Analysis
Tools	Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab

1. Project Overview

This project was completed as part of my Data Analyst Internship at Alfido Tech. It analyses a retail Superstore sales dataset to understand how sales are distributed across regions, product categories, individual products, customer segments and time.

Sales performance analysis is useful because raw transaction records on their own do not tell a business very much. By grouping and summarising the data, it becomes possible to see which regions and products are generating the most revenue, which ones are underperforming, and how demand changes month by month. This kind of information supports decisions about inventory planning, marketing budgets and where to focus sales effort.

Objective

The objective stated in the notebook was to analyse sales performance across regions, categories, products and time; to identify the best and worst performing products; to understand seasonality; and to provide actionable recommendations for improving sales.

Headline Figures

KPI	Value
Total Revenue	\$2,261,536.78
Total Orders	4,922
Total Customers	793
Average Order Value	\$459.48

A conversion rate was deliberately not calculated. The dataset contains no website visits, sessions or leads, so there is no denominator for a conversion funnel. Total revenue, total orders, total customers and average order value were used as the main KPIs instead.

2. Dataset Overview

The dataset used is *superstore_final_dataset (1).csv*, loaded into Google Colab with Pandas using latin-1 encoding. In its original form it contains **9,800 rows and 18 columns**. After creating additional date-based columns during preparation, the working dataset has 9,800 rows and 23 columns.

Property	Value
Rows	9,800
Columns (original)	18
Columns (after feature creation)	23
Unique orders	4,922
Unique customers	793
Unique products	1,849
Regions	4
Categories	3
Sub-categories	17

Property	Value
States	49
Cities	529
Country	United States (single country)
Order date range	3 January 2015 to 30 December 2018

Columns and Data Types

Column	Type	Description
Row_ID	int64	Row number
Order_ID	object	Unique order identifier
Order_Date	datetime64	Date the order was placed
Ship_Date	datetime64	Date the order was shipped
Ship_Mode	object	Shipping method (4 values)
Customer_ID / Customer_Name	object	Customer identifier and name
Segment	object	Consumer, Corporate, Home Office
Country / City / State	object	Location of the customer
Postal_Code	float64	Postal code (11 values missing)
Region	object	West, East, Central, South
Product_ID / Product_Name	object	Product identifier and name
Category	object	Technology, Furniture, Office Supplies
Sub_Category	object	17 sub-categories such as Phones, Chairs
Sales	float64	Sales value of the line item

Sales is the key numeric variable. Its mean is 230.77 with a standard deviation of 626.65, a minimum of 0.44 and a maximum of 22,638.48. The median is only 54.49, which shows that most line items are small and a few very large transactions pull the average upwards.

3. Data Cleaning and Preparation

The following steps were actually carried out in the notebook:

- **Missing value check.** `isnull().sum()` was used on every column. Only **Postal_Code** had missing values – 11 rows, which is 0.11% of the data.
- **Inspecting the missing rows.** All 11 rows were found to belong to Burlington, Vermont, in the East region. They were kept, because they contain valid sales and order information and **Postal_Code** is not used anywhere in the sales analysis. All 9,800 rows were therefore retained.
- **Duplicate check.** `duplicated().sum()` returned 0, and `drop_duplicates()` was run as a confirmation step. The count remained 0, so no rows were lost.
- **Date conversion.** **Order_Date** and **Ship_Date** were stored as text. Both were converted to datetime using `pd.to_datetime(..., dayfirst=True)`, since the dates are written in day-first format.

- **Feature creation.** Five new columns were derived from Order_Date: **Year**, **Month**, **Month_Name**, **Quarter** and **Year_Month**. These made the time-based analysis possible and increased the column count from 18 to 23.
- **Structure checks.** *shape*, *info()*, *dtypes* and *describe()* were used to confirm the data types were correct after cleaning.

4. Exploratory Data Analysis

The analysis was carried out mainly with Pandas *groupby* aggregations and one pivot table. The areas covered were:

4.1 Overall Performance

Total revenue across the four years is \$2,261,536.78 from 4,922 orders placed by 793 customers, giving an average order value of \$459.48.

4.2 Regional and State Performance

Sales were grouped by Region and by State. The West leads with \$710,219.68 and the South is last with \$389,151.46. At state level, California alone contributes \$446,306.46 – more than the entire South region.

4.3 Category and Sub-Category Performance

The three categories are close together, with Technology highest at \$827,455.87 and Office Supplies lowest at \$705,422.33. At sub-category level the spread is much wider: Phones leads with \$327,782.45 while Fasteners generates only \$3,001.96.

4.4 Product Performance

Products were ranked by total sales, and a combined table of total sales, number of orders and number of customers was also produced for the top products. The results show that high revenue does not always mean high order volume.

Product (top 5 by sales)	Total Sales	Orders	Customers
Canon imageCLASS 2200 Advanced Copier	61,599.82	5	5
Fellowes PB500 Electric Punch Plastic Comb Binding Machine	27,453.38	10	10
Cisco TelePresence System EX90 Videoconferencing Unit	22,638.48	1	1
HON 5400 Series Task Chairs for Big and Tall	21,870.58	8	8
GBC DocuBind TL300 Electric Binding System	19,823.48	11	11

The Cisco TelePresence unit reached \$22,638.48 from a single order, which is also the largest single Sales value in the whole dataset.

4.5 Time-Based Trends

Sales were aggregated by year, by year-month and by calendar month. Yearly totals were \$479,856.21 (2015), \$459,436.01 (2016), \$600,192.55 (2017) and \$722,052.02 (2018) – a dip in 2016 followed by two years of growth. Grouping by calendar month across all four years revealed a clear seasonal pattern.

4.6 Customer Segments

Sales were grouped by Segment. Consumer is the largest segment at roughly \$1,148,061, followed by Corporate at about \$688,494 and Home Office at about \$424,982.

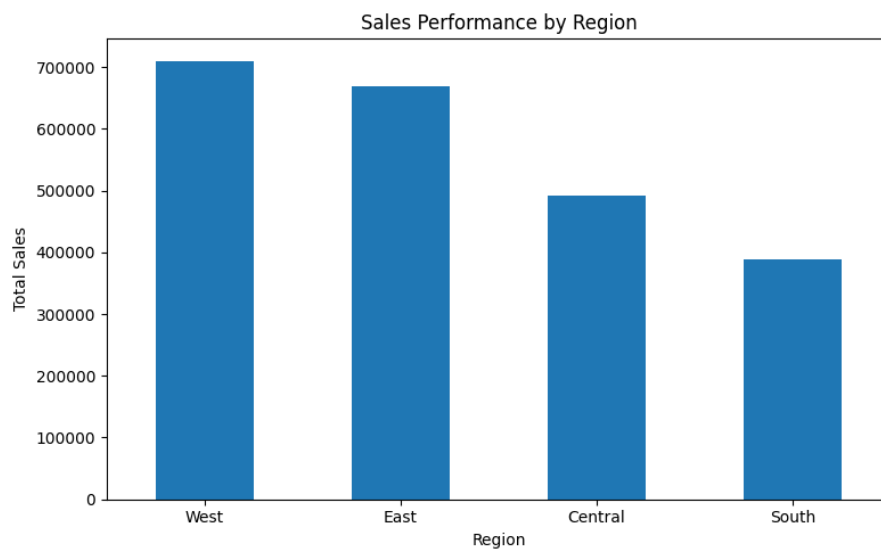
4.7 Region and Category Combined

A pivot table of Region against Category was created and displayed as a heatmap to see which combinations perform best.

5. Visualizations

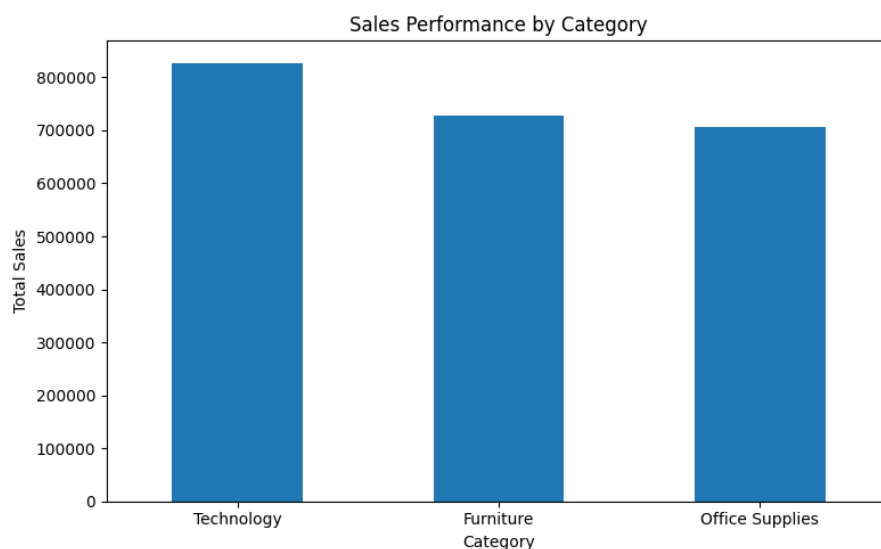
All charts below are the actual outputs produced in the notebook using Matplotlib and Seaborn.

Figure 1 – Sales Performance by Region

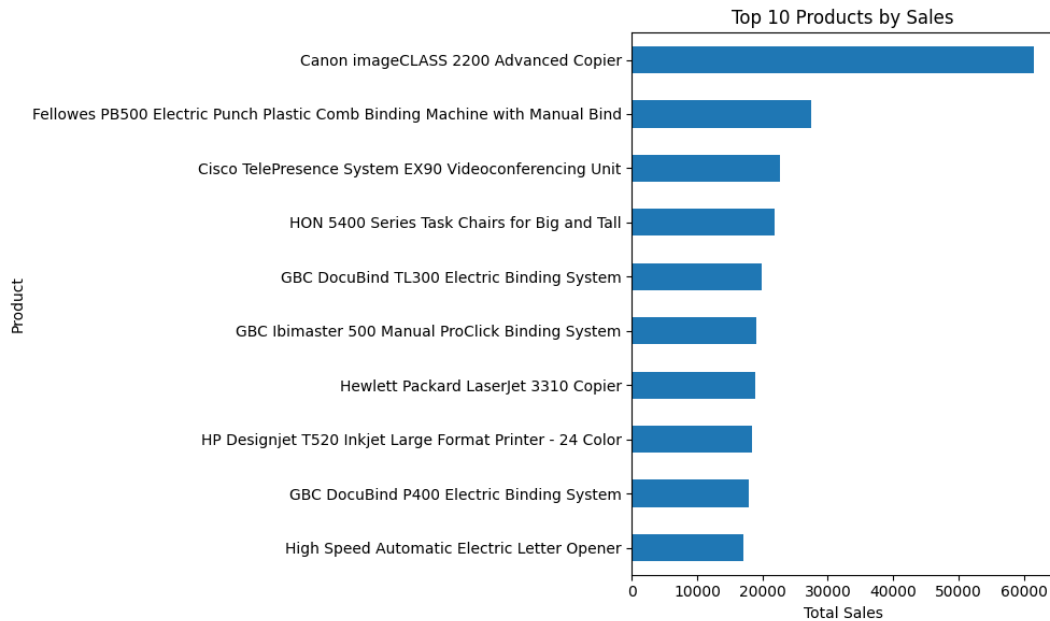


The bar chart ranks the four regions by total sales. West is highest at \$710,219.68, followed by East at \$669,518.73. Central (\$492,646.91) and South (\$389,151.46) are clearly behind – the South earns only about 55% of what the West earns.

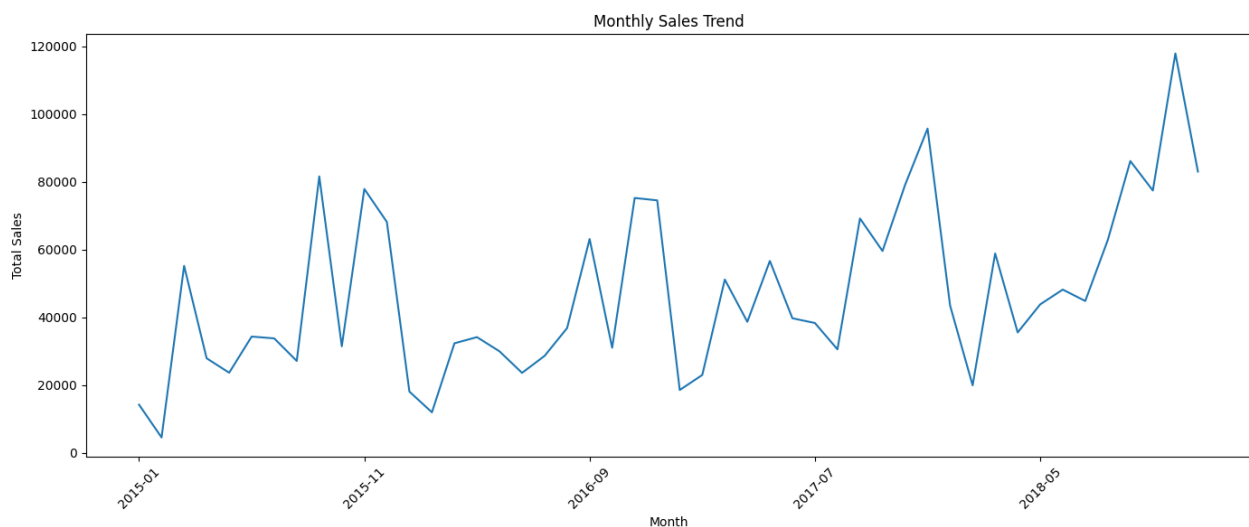
Figure 2 – Sales Performance by Category



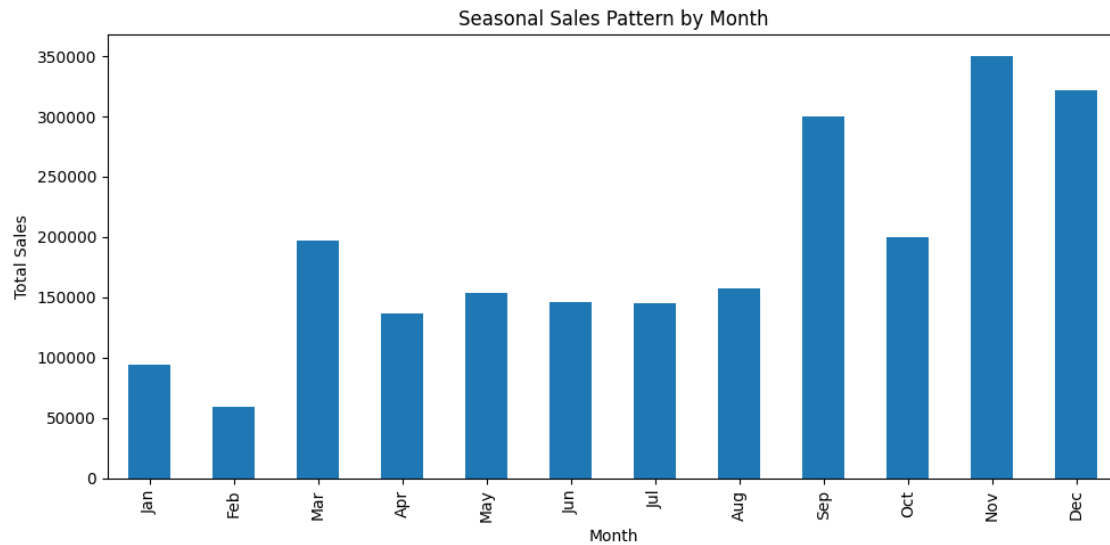
Technology leads with \$827,455.87, Furniture follows with \$728,658.58 and Office Supplies is lowest with \$705,422.33. The gap between the highest and lowest category is only about \$122,000, so revenue is spread fairly evenly across categories.

Figure 3 – Top 10 Products by Sales

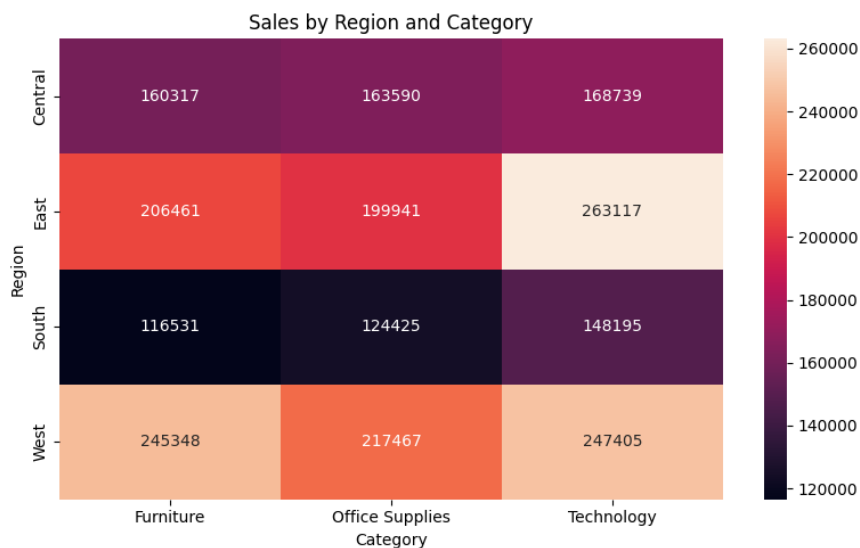
The Canon imageCLASS 2200 Advanced Copier is far ahead of every other product at \$61,599.82 – more than twice the second product, the Fellowes PB500 binding machine at \$27,453.38. Most of the top ten are office machines, copiers and binding systems.

Figure 4 – Monthly Sales Trend (2015–2018)

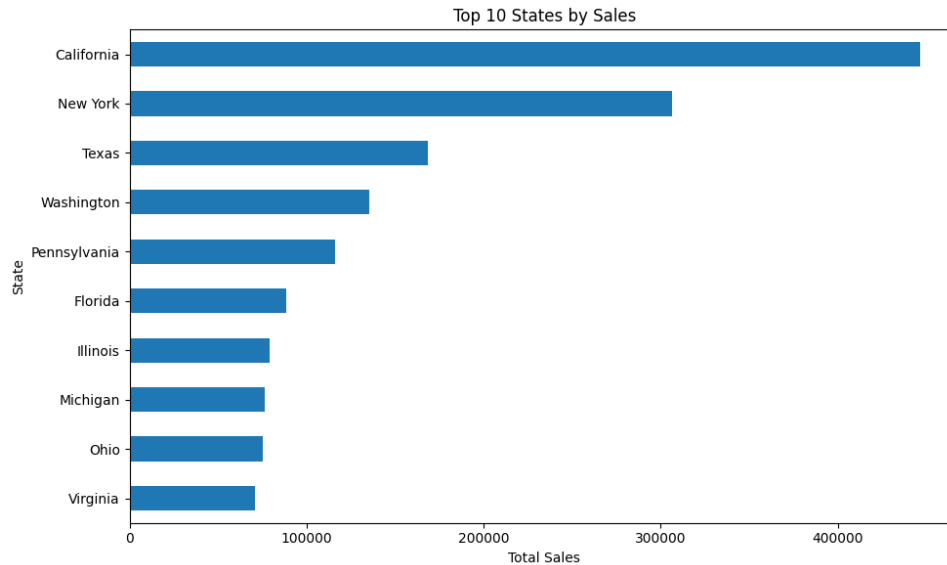
The line chart plots total sales for each year-month from January 2015 to December 2018. The series is very uneven from month to month, but the peaks become higher towards the end of the period, matching the rise in yearly totals from \$479,856.21 in 2015 to \$722,052.02 in 2018.

Figure 5 – Seasonal Sales Pattern by Month

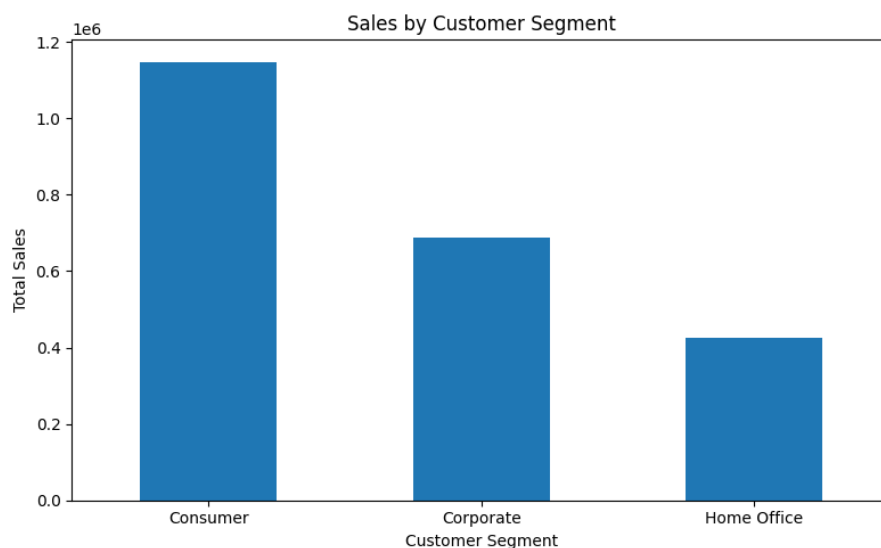
Sales for each calendar month are combined across all four years. November is the strongest month at \$350,161.71, followed by December (\$321,480.17) and September (\$300,103.41). February is the weakest at \$59,371.12 – less than a sixth of November. The last four months of the year clearly carry the business.

Figure 6 – Sales by Region and Category (Heatmap)

The heatmap shows the pivot table of Region against Category. The single strongest combination is East – Technology at \$263,117, followed by West – Technology at \$247,405 and West – Furniture at \$245,348. The weakest is South – Furniture at \$116,531.

Figure 7 – Top 10 States by Sales

California dominates with \$446,306.46, ahead of New York at \$306,361.15 and Texas at \$168,572.53. Sales are heavily concentrated: the top two states together account for a large share of the \$2.26 million total.

Figure 8 – Sales by Customer Segment

The Consumer segment is the largest at roughly \$1,148,061, which is more than Corporate (about \$688,494) and Home Office (about \$424,982) individually. Consumer alone accounts for roughly half of all revenue.

6. Key Findings

- **Best performing region:** West, with \$710,219.68 in sales. **Lowest performing region:** South, with \$389,151.46.
- **Best performing category:** Technology, with \$827,455.87. **Lowest performing category:** Office Supplies, with \$705,422.33. The three categories are reasonably balanced.
- **Best performing product:** Canon imageCLASS 2200 Advanced Copier, with \$61,599.82 from only 5 orders. **Lowest performing product:** Eureka Disposable Bags for Sanitaire Vibra Groomer I Upright Vac, with just \$1.62.
- **Strongest sub-category:** Phones (\$327,782.45), closely followed by Chairs (\$322,822.73). **Weakest sub-category:** Fasteners (\$3,001.96).

- **Strongest state:** California (\$446,306.46), which on its own exceeds the total sales of the whole South region.
- **Sales trend:** Yearly sales dipped from \$479,856.21 in 2015 to \$459,436.01 in 2016, then grew to \$600,192.55 in 2017 and \$722,052.02 in 2018.
- **Seasonality:** November (\$350,161.71), December (\$321,480.17) and September (\$300,103.41) are the strongest months; February (\$59,371.12) and January (\$94,291.63) are the weakest.
- **Customer segments:** Consumer is the largest contributor at roughly \$1,148,061, about half of total revenue.
- **Unusual pattern:** High revenue does not always mean high demand. The Cisco TelePresence System EX90 reached \$22,638.48 from a single order, while the median line-item sale is only \$54.49. A small number of expensive transactions has a large effect on the totals.

7. Business Recommendations

- **Investigate and support the South region.** The South generates \$389,151.46 against the West's \$710,219.68 and is the weakest in every category. Understanding the reason for this gap should be a priority before spending more on already strong regions.
- **Plan stock and campaigns around the September–December peak.** These four months produce the highest sales of the year, with November at \$350,161.71. Inventory and promotions should be prepared in advance of this window, and the quiet January–February period could be used for clearance or off-season offers.
- **Protect the high-value product lines.** Copiers, binding machines and other office equipment fill most of the top ten, with the Canon imageCLASS 2200 alone contributing \$61,599.82. Keeping these items in stock and well presented protects a meaningful share of revenue, though the small order counts mean the business should not rely on them alone.
- **Review the weakest products and sub-categories.** Items such as the Eureka disposable bags (\$1.62) and the Fasteners sub-category (\$3,001.96) contribute very little. Their pricing, visibility and shelf space should be reviewed, and discontinued lines removed.
- **Target the Consumer segment while developing the others.** Consumer contributes roughly \$1,148,061, about half of all revenue. Loyalty offers suit this group, while Corporate (about \$688,494) and Home Office (about \$424,982) represent clear room for growth through bulk or business-focused offers.
- **Build on the strongest region–category pairs.** East–Technology (\$263,117) and West–Technology (\$247,405) are the two strongest combinations and are sensible places to concentrate technology marketing.

8. Conclusion

This project examined 9,800 sales records covering four years, \$2,261,536.78 of revenue and 4,922 orders. Working through the data showed that a fairly simple set of Pandas groupings can answer real business questions: which region is falling behind, when demand peaks, which products matter most and which contribute almost nothing.

The findings can support practical decisions. Knowing that November and December are the strongest months helps with inventory planning; knowing that the South underperforms across every category tells management where to investigate; knowing that the Consumer segment provides about

half of revenue helps direct marketing spend.

The project also gave me useful hands-on practice. I handled a real dataset with genuine imperfections – text-formatted dates and missing postal codes – decided how to treat them and justified those decisions rather than removing data automatically. I also learned that being honest about what the data cannot show matters: a conversion rate could not be calculated here, so it was left out instead of being invented.

9. Tools and Technologies

Tool	How it was used
Python	Core programming language for the whole analysis
Pandas	Loading the CSV, cleaning, groupby aggregations, pivot table
NumPy	Imported alongside Pandas for numeric support
Matplotlib	Bar charts, horizontal bar charts and the line trend chart
Seaborn	Heatmap of sales by region and category
Google Colab	Notebook environment used to write and run the analysis

10. Project Skills Demonstrated

- **Data cleaning and preparation** – missing value analysis, duplicate checking, date type conversion and reasoned decisions about which rows to keep.
- **Feature engineering** – deriving Year, Month, Month_Name, Quarter and Year_Month columns from the order date to enable time-based analysis.
- **Exploratory data analysis** – groupby aggregations, ranking, a pivot table and descriptive statistics across several business dimensions.
- **Data visualization** – building bar charts, horizontal bar charts, a line chart and a heatmap with clear titles and axis labels using Matplotlib and Seaborn.
- **Python and Pandas** – practical use of Pandas for real analysis rather than exercises.
- **Business insight generation** – turning aggregated numbers into findings and recommendations that a business could act on.
- **Analytical honesty** – recognising that conversion rate could not be measured from this dataset and documenting that instead of producing a misleading metric.